Official Benchmark Dossier

Читаемая версия для экспертов: структура, KPI, таблицы и источники.
Readable expert edition: structure, KPIs, analytical tables and official sources.

generated_utc=2026-04-06T00:23:22Z | runs=20260405T230026Z_0b28f298, 20260405T230137Z_5d7213c1, 20260405T162225Z_22f14610, 20260405T230146Z_5dbac053, 20260406T000140Z_669d8f63 | tracks=ann, core, online | tiers=desktop, edge, server | scales=100M, 10M, 1M

`overall_status`

`pass`

`rows_total`

`289`

`rows_ok`

`289`

1. Gate Status

Gate	Value
overall_status	pass
correctness	pass
reproducibility	pass
investor	pass
scale	pass
final_dossier_ready	True
ok_scales	100M, 10M, 1M

2. Executive KPIs

> Top throughput and memory-efficiency points for quick specialist review.

2.1 Top Throughput

track	dataset	tier	scale	algorithm	qps	p95(ms)
core	core:adversarial	desktop	1M	core256/match_batch	403998779.924	0.000002
core	core:adversarial	server	1M	core256/match_batch	395728506.501	0.000003
core	core:adversarial	server	1M	core256/match_batch	389958566.902	0.000003
core	core:adversarial	desktop	1M	core256/match_batch	286875722.568	0.000003
core	core:adversarial	server	1M	core256/match_batch	255004462.578	0.000004
core	core:adversarial	desktop	1M	core256/match_batch	254215205.883	0.000004
core	core:adversarial	edge	1M	core256/match_batch	252729478.366	0.000004
core	core:adversarial	edge	1M	core256/match_batch	252495284.651	0.000004
core	core:adversarial	server	1M	core256/match_batch	251954854.729	0.000004
core	core:adversarial	edge	1M	core256/match_batch	251608406.740	0.000004
core	core:adversaria	desktop	1M	core256/match_ba	251330796.568	0.000004

track	dataset	tier	scale	algorithm	qps	p95(ms)
	l			tch		
core	core:adversarial	server	1M	core256/match_batch	250580720.821	0.000004
core	core:adversarial	edge	1M	core256/match_batch	250314144.251	0.000004
core	core:adversarial	desktop	1M	core256/match_batch	250133508.760	0.000004
core	core:adversarial	edge	1M	core256/match_batch	247692128.592	0.000004
core	core:clustered	desktop	1M	core256/match_batch	162192189.635	0.000006
core	core:clustered	server	1M	core256/match_batch	136984987.815	0.000007
core	core:clustered	server	1M	core256/match_batch	135182266.250	0.000007
core	core:clustered	desktop	1M	core256/match_batch	134928639.616	0.000007
core	core:clustered	edge	1M	core256/match_batch	132234639.790	0.000008

2.2 Top Memory Efficiency

track	dataset	tier	scale	algorithm	qps/indexMB	qps
core	core:adversaria	desktop	1M	core256/match_ba	1409897440.816	403998779924

track	dataset	tier	scale	algorithm	qps/indexMB	qps
	l			tch		
core	core:adversarial	server	1M	core256/match_batch	1381035380.054	395728506501
core	core:adversarial	server	1M	core256/match_batch	1360899123.516	389958566902
core	core:adversarial	desktop	1M	core256/match_batch	1001154872.689	286875722568
core	core:adversarial	server	1M	core256/match_batch	889928774.669	255004462578
core	core:adversarial	desktop	1M	core256/match_batch	887174382.700	254215205883
core	core:adversarial	edge	1M	core256/match_batch	881989408.074	252729478366
core	core:adversarial	edge	1M	core256/match_batch	881172105.802	252495284651
core	core:adversarial	server	1M	core256/match_batch	879286083.366	251954854729
core	core:adversarial	edge	1M	core256/match_batch	878077029.880	251608406740
core	core:adversarial	desktop	1M	core256/match_batch	877108210.441	251330796568
core	core:adversarial	server	1M	core256/match_batch	874490554.326	250580720821
core	core:adversaria	edge	1M	core256/match_ba	873560240.568	250314144251

track	dataset	tier	scale	algorithm	qps/indexMB	qps
	l			tch		
core	core:adversarial	desktop	1M	core256/match_batch	872929848.773	250133508760
core	core:adversarial	edge	1M	core256/match_batch	864409784.301	247692128592
core	core:adversarial	server	1M	core256/probe	114086308.713	32690838.491
core	core:adversarial	server	1M	core256/predict	113962900.109	32655476.397
core	core:adversarial	desktop	1M	core256/probe	113423261.761	32500845.835
core	core:adversarial	edge	1M	core256/predict	111815285.361	32040088.559
core	core:adversarial	desktop	1M	core256/probe	111241965.579	31875806.757

3. Data Impact Analysis

track	dataset	regime	tier	scale	mean_qps
ann	openml:adult	mixed tabular (numeric + categorical)	desktop	1M	23887.930
ann	openml:adult	mixed tabular (numeric + categorical)	edge	1M	19542.167
ann	openml:adult	mixed tabular (numeric +	server	100	25832.586

track	dataset	regime	tier	scale	mean_qps
		categorical)		M	
ann	openml:adult	mixed tabular (numeric + categorical)	server	10M	19370.696
ann	openml:adult	mixed tabular (numeric + categorical)	server	1M	21307.299
ann	synthetic_binary256	binary clustered synthetic	desktop	1M	7571.491
ann	synthetic_binary256	binary clustered synthetic	edge	1M	6764.483
ann	synthetic_binary256	binary clustered synthetic	server	100M	8990.447
ann	synthetic_binary256	binary clustered synthetic	server	10M	7439.561
ann	synthetic_binary256	binary clustered synthetic	server	1M	7644.110
core	core:adversarial	prefix-collision adversarial	desktop	1M	88383149.805
core	core:adversarial	prefix-collision adversarial	edge	1M	78556331.732
core	core:adversarial	prefix-collision adversarial	server	1M	92930428.519
core	core:clustered	near-duplicate clustered keys	desktop	1M	43269405.948
core	core:clustered	near-duplicate clustered keys	edge	1M	41182419.026
core	core:clustered	near-duplicate clustered keys	server	1M	41105757.775
core	core:legacy_compat	core-legacy_compat	desktop	1M	93000.000

track	dataset	regime	tier	scale	mean_qps
core	core:legacy_compat	core-legacy_compat	edge	1M	111000.000
core	core:legacy_compat	core-legacy_compat	server	1M	91000.000
core	core:random	uniform-random 256-bit	desktop	1M	35224856.627
core	core:random	uniform-random 256-bit	edge	1M	34499440.951
core	core:random	uniform-random 256-bit	server	1M	33629342.445
core	core:replay	replay/hash-stream	desktop	1M	35252122.368
core	core:replay	replay/hash-stream	edge	1M	35333782.000
core	core:replay	replay/hash-stream	server	1M	35086573.792
online	online:synthetic_drift	stream with concept drift	desktop	1M	34593.010
online	online:synthetic_drift	stream with concept drift	edge	1M	35093.895
online	online:synthetic_drift	stream with concept drift	server	1M	38040.678

4. Summary Table (Top by QPS)

dataset	tier	scale	algorithm	track	status	qps	pm
core:adversarial	server	1M	core256/match_batch	core	ok	308645422.306	0.00
core:adversarial	desktop	1M	core256/match_ba	core	ok	289310802.741	0.00

dataset	tier	scale	algorithm	track	status	qps	ps m
			tch				
core:adversarial	edge	1M	core256/match_batch	core	ok	250967888.520	0.00
core:clustered	desktop	1M	core256/match_batch	core	ok	120199661.410	0.00
core:clustered	server	1M	core256/match_batch	core	ok	113008934.367	0.00
core:clustered	edge	1M	core256/match_batch	core	ok	112580448.836	0.00
core:replay	server	1M	core256/match_batch	core	ok	92325513.669	0.01
core:replay	edge	1M	core256/match_batch	core	ok	91377371.668	0.01
core:random	desktop	1M	core256/match_batch	core	ok	91194071.564	0.01
core:replay	desktop	1M	core256/match_batch	core	ok	90886101.328	0.01
core:random	edge	1M	core256/match_batch	core	ok	88225160.582	0.01
core:random	server	1M	core256/match_batch	core	ok	86350424.310	0.01
core:adversarial	desktop	1M	core256/probe	core	ok	30807204.066	0.01
core:adversaria	edge	1M	core256/predict	core	ok	30570451.5	0.

dataset	tier	scale	algorithm	track	status	qps	psm
l						34	01
core:adversarial	desktop	1M	core256/predict	core	ok	29983608.210	0.02
core:adversarial	server	1M	core256/probe	core	ok	29856975.882	0.02
core:adversarial	server	1M	core256/predict	core	ok	29728949.060	0.01
core:adversarial	edge	1M	core256/probe	core	ok	29323738.462	0.02
core:clustered	desktop	1M	core256/probe	core	ok	25850241.770	0.03
core:clustered	desktop	1M	core256/predict	core	ok	25784922.090	0.03
core:clustered	edge	1M	core256/predict	core	ok	25648950.755	0.03
core:clustered	server	1M	core256/probe	core	ok	25591883.795	0.03
core:clustered	edge	1M	core256/probe	core	ok	25291012.125	0.03
core:replay	desktop	1M	core256/predict	core	ok	24688315.038	0.03
core:clustered	server	1M	core256/predict	core	ok	24619010.965	0.03

5. External Official Benchmarks

category	benchmark	scope	url
ANN / Similarity	ANN-Benchmarks	Recall-latency-memory tradeoffs across ANN	https://github.com/erikbern/ann-benchmarks

category	benchmark	scope	url
		implementations	
ANN / Similarity	ANN-Benchmarks website plots	Recall/QPS, Recall/Build time, Recall/Index size, latency percentiles	https://ann-benchmarks.com/
ANN / Similarity	Big-ANN (NeurIPS track)	Large-scale practical vector search; normalized hardware, recall/QPS leaderboards	https://big-ann-benchmarks.com/neurips23.html
ANN / Binary	Faiss Binary Hashing Benchmark	256-bit binary vectors; recall, wall-clock, distance computations, random accesses	https://github.com/facebookresearch/faiss/wiki/Binary-hashing-index-benchmark
Tree / KV serving	YCSB	Read/write/update/scan workloads with Zipfian and related distributions	https://github.com/brianfrankcooper/YCSB
Tree / KV serving	YCSB workload A	50/50 read-update, Zipfian request distribution	https://raw.githubusercontent.com/brianfrankcooper/YCSB/master/workloads/workloada
Tree / KV serving	YCSB workload F	50/50 read-read-modify-write, Zipfian request distribution	https://raw.githubusercontent.com/brianfrankcooper/YCSB/master/workloads/workloadf
Tree / Storage engine	RocksDB db_bench	Official benchmark tool with fill/read/seek/mixed workloads	https://github.com/facebook/rocksdb/wiki/Benchmarking-tools
Tree / Sorted index	SOSD (Search on Sorted Data)	Sorted-data lookup benchmark with many datasets and baseline index structures	https://github.com/learnedsystems/SOSD
Truth-table / Logic compression	EPFL Combinational Benchmark Suite	Combinational logic suite for synthesis/optimization (arithmetic/control/	https://github.com/lsils/benchmarks

category	benchmark	scope	url
		MtM)	
Truth-table / Logic compression	Berkeley multi-output PLA tables (table_ex)	PLA table benchmark package used in cube-hashing extraction work	https://people.eecs.berkeley.edu/~alanmi/benchmarks/table_ex/
Truth-table / Logic compression	IWLS 2005 benchmark bundle	Widely used logic synthesis benchmark set (AIG/Verilog archives)	https://people.eecs.berkeley.edu/~alanmi/benchmarks/iwls2005/

6. Notes

- This readable dossier is generated from machine-validated CSV/JSON artifacts in the same output folder.
- For reproducibility/audit, use `OFFICIAL_STACK/05_REPRO_LEGAL`.
- For full raw points, use `report_summary.csv` and `dossier_evidence_matrix.csv`.